A Proof of Theorem 5.2 (NCER Noise Reduction)

Proof. Decompose the observed graph as $\boldsymbol{S} = \boldsymbol{S}^* + \boldsymbol{N}$. The NCER reweighting applies element-wise multiplication with $\boldsymbol{W}$ where $W_{ij} = 1 + \kappa \cdot c_{ij}$:

$$\hat{\boldsymbol{S}} = \boldsymbol{S} \circ \boldsymbol{W} = (\boldsymbol{S}^* + \boldsymbol{N}) \circ \boldsymbol{W} \tag{20}$$

$$= \boldsymbol{S}^* \circ \boldsymbol{W} + \boldsymbol{N} \circ \boldsymbol{W} = \hat{\boldsymbol{S}}^* + \hat{\boldsymbol{N}}. \tag{21}$$

Consider the Frobenius norms. For any matrix $\boldsymbol{M}$ with element-wise reweighting:

$$\|\boldsymbol{M} \circ \boldsymbol{W}\|_F^2 = \sum_{(i,j):M_{ij}\neq 0} M_{ij}^2 (1 + \kappa c_{ij})^2. \tag{22}$$

For the noise component, define the average consistency on noise edges:

$$\bar{c}_N = \frac{\sum_{(i,j)\in \mathrm{supp}(\boldsymbol{N})} N_{ij}^2 c_{ij}}{\sum_{(i,j)\in \mathrm{supp}(\boldsymbol{N})} N_{ij}^2}. \tag{23}$$

By the Cauchy-Schwarz inequality applied to the weighted sum:

$$\|\hat{\boldsymbol{N}}\|_F^2 \leq \|\boldsymbol{N}\|_F^2 \cdot (1 + \kappa \bar{c}_N)^2 + O(\kappa^2 \mathrm{Var}(c_N)), \tag{24}$$

where the second-order term accounts for variance in consistency scores. Similarly for the signal:

$$\|\hat{\boldsymbol{S}}^*\|_F^2 \geq \|\boldsymbol{S}^*\|_F^2 \cdot (1 + \kappa \bar{c}_{S^*})^2. \tag{25}$$

Therefore, the noise-to-signal ratio satisfies:

$$\frac{\|\hat{\boldsymbol{N}}\|_F}{\|\hat{\boldsymbol{S}}^*\|_F} \leq \frac{\|\boldsymbol{N}\|_F}{\|\boldsymbol{S}^*\|_F} \cdot \frac{1 + \kappa \bar{c}_N}{1 + \kappa \bar{c}_{S^*}} + O(\kappa^2). \tag{26}$$

Under the community structure assumption (Assumption 5.1), clean edges connect items within the same semantic community and thus share many neighbors ($\bar{c}_{S^*}$ is large), while noise edges connect items across communities with few shared neighbors ($\bar{c}_N$ is small). Therefore $\bar{c}_{S^*} > \bar{c}_N$, and the multiplicative factor $(1 + \kappa \bar{c}_N)/(1 + \kappa \bar{c}_{S^*}) < 1$ for all $\kappa > 0$.

Extending from $\boldsymbol{S}^*$ to $\boldsymbol{S}$ in the denominator introduces an additional multiplicative correction bounded by $(1 + \epsilon/\|\boldsymbol{S}^*\|_F)^{-1}$, yielding the stated bound. □

B Proof of Theorem 5.3 (RIG Non-Degeneracy)

Proof. The BPR loss for a triplet (u, p, n) is:

$$\mathcal{L}_{u,p,n} = -\log \sigma(\boldsymbol{e}_u^\top \boldsymbol{e}_p - \boldsymbol{e}_u^\top \boldsymbol{e}_n). \tag{27}$$

With RIG, $\boldsymbol{e}_i = \boldsymbol{e}_i^{\mathrm{UI}} + g_i \cdot \boldsymbol{h}_i$. Taking the derivative with respect to g_p (gate of positive item):

$$\frac{\partial \mathcal{L}_{u,p,n}}{\partial g_p} = -\sigma(-\Delta_{u,p,n}) \cdot \boldsymbol{e}_u^\top \frac{\partial \boldsymbol{e}_p}{\partial g_p} \tag{28}$$

$$= -\sigma(-\Delta_{u,p,n}) \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_p, \tag{29}$$

where $\Delta_{u,p,n} = \boldsymbol{e}_u^\top \boldsymbol{e}_p - \boldsymbol{e}_u^\top \boldsymbol{e}_n$ and $\sigma(-\Delta) = 1 - \sigma(\Delta) \in (0, 1)$.

Define $w_{u,p} = \sigma(-\Delta_{u,p,n}) > 0$. Aggregating over all triplets where i appears as positive:

$$\left.\frac{\partial \mathcal{L}}{\partial g_i}\right|_{\text{positive}} = -\sum_{u:(u,i,\cdot)\in\mathcal{D}} w_{u,i} \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_i. \tag{30}$$

Similarly, when i appears as a *negative* item in triplet (u, p, i):

$$\frac{\partial \mathcal{L}_{u,p,i}}{\partial g_i} = -\sigma(-\Delta_{u,p,i}) \cdot \boldsymbol{e}_u^\top \left(-\frac{\partial \boldsymbol{e}_i}{\partial g_i}\right) \tag{31}$$

$$= \sigma(-\Delta_{u,p,i}) \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_i. \tag{32}$$

Define $w'_{u,i} = \sigma(-\Delta_{u,p,i}) > 0$. Aggregating:

$$\left.\frac{\partial \mathcal{L}}{\partial g_i}\right|_{\text{negative}} = \sum_{u:(u,\cdot,i)\in\mathcal{D}} w'_{u,i} \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_i. \tag{33}$$

At a stationary point, $\partial\mathcal{L}/\partial g_i = 0$, which requires:

$$\sum_{u\in\mathrm{pos}} w_{u,i} \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_i = \sum_{u\in\mathrm{neg}} w'_{u,i} \cdot \boldsymbol{e}_u^\top \boldsymbol{h}_i. \tag{34}$$

Case 1: $\boldsymbol{e}_u^\top \boldsymbol{h}_i > 0$ on average (helpful II signal). The positive-side gradient pushes g_i upward (increasing g_i reduces $\mathcal{L}$), while the negative-side gradient pushes g_i upward as well (increasing g_i increases the negative item's score, which increases $\mathcal{L}$). The balance is achieved at an interior point because the BPR weights w, w' depend on the margin: as g_i increases, the margins change, adjusting the weights until equilibrium.

Case 2: $\boldsymbol{e}_u^\top \boldsymbol{h}_i < 0$ on average (harmful II signal). The roles reverse, and the equilibrium g_i is pushed toward lower values.

Since the sigmoid output of the gate MLP is continuous and strictly bounded in $(0, 1)$ (assuming bounded inputs and non-degenerate initialization), gradient descent converges to $g_i^* \in (0, 1)$ that balances the positive and negative contributions. Furthermore, if there exist items i, j with $\mathbb{E}[\boldsymbol{e}_u^\top \boldsymbol{h}_i] > 0 > \mathbb{E}[\boldsymbol{e}_u^\top \boldsymbol{h}_j]$, the equilibrium conditions require $g_i^* > g_j^*$. □

C Proof of Theorem 5.5 (INA Generalization Bound)

Proof. The standard PAC-Bayes bound for BPR with n observed interactions states:

$$R(f) \leq \hat{R}_n(f) + \sqrt{\frac{\mathrm{KL}(f\|\pi) + \log(2\sqrt{n}/\delta)}{2n}}, \tag{35}$$

with probability at least $1 - \delta$ over the draw of the training set.

INA augments the training set with $n_{\mathrm{aug}} = n \cdot 1$ additional soft-positive triplets (one per observed interaction). Each augmented triplet (u, j, n') satisfies $j \in \mathcal{N}_{\mathrm{II}}(p)$ where (u, p) is a true interaction.

The key question is the *effective sample size* contributed by augmented samples. Define the *soft label quality*:

$$q = \mathbb{P}[\text{user } u \text{ prefers } j \text{ over random } n' | j \in \mathcal{N}_{\mathrm{II}}(p), (u, p) \in \boldsymbol{R}^+]. \tag{36}$$

This is the probability that a neighbor of a positive item is itself a relevant item. The discount factor δ in INA accounts for this uncertainty. Following the analysis of learning with noisy labels [11], each augmented sample contributes an effective fraction $\delta_{\mathrm{eff}} = \delta^2 \cdot q$ to the sample size. This arises because the discount δ reduces both the gradient magnitude (linear in δ) and the effective signal strength (linear in $\delta \cdot q$).

Substituting the effective sample size $n + \delta_{\mathrm{eff}} \cdot n_{\mathrm{aug}}$ into the PAC-Bayes bound:

$$R(f) \leq \hat{R}_{n+n_{\mathrm{aug}}}(f) + \sqrt{\frac{\mathrm{KL}(f\|\pi) + \log(2\sqrt{n + \delta_{\mathrm{eff}} n_{\mathrm{aug}}}/\delta)}{2(n + \delta_{\mathrm{eff}} n_{\mathrm{aug}})}}. \tag{37}$$

When the II graph has positive predictive value ($q > 0$) and the discount is positive ($\delta > 0$), we have $\delta_{\mathrm{eff}} > 0$ and thus $n + \delta_{\mathrm{eff}} n_{\mathrm{aug}} > n$, yielding a strictly tighter bound than standard BPR. □

Algorithm 1: IIMRec Training Procedure

Require: Interaction matrix R, multimodal features $\{F^m\}$

Preprocessing (one-time):
Construct semantic graph S_{sem} (spatial + spectral KNN)
Construct co-occurrence graph S_{co} from $R^\top R$
Fuse: $S_{\text{fused}} = S_{\text{co}} + S_{\text{sem}}$
Apply NCER reweighting (Eq. (8)) $\rightarrow S_{\text{NCER}}$
Build content-guided UI graph $\tilde{A}$
Pre-compute INA neighbor table from S_{NCER}
for epoch = 1 to max_epochs **do**
  for each mini-batch $(u, p, n) \in \mathcal{D}$ **do**
    Project modality features (Eq. (1))
    Propagate on UI graph (Eq. (9))
    Fuse user modality representations (Eq. (10))
    Propagate item representations on II graph
    Apply RIG gating (Eq. (12))
    Calculate contrastive learning losses (Eq. (15))
    Compute $\mathcal{L}_{\text{BPR}} + \lambda_{\text{INA}}\mathcal{L}_{\text{INA}} + \lambda_{\text{CL}}\mathcal{L}_{\text{CL}}$
    Update parameters via backpropagation
  end for
end for

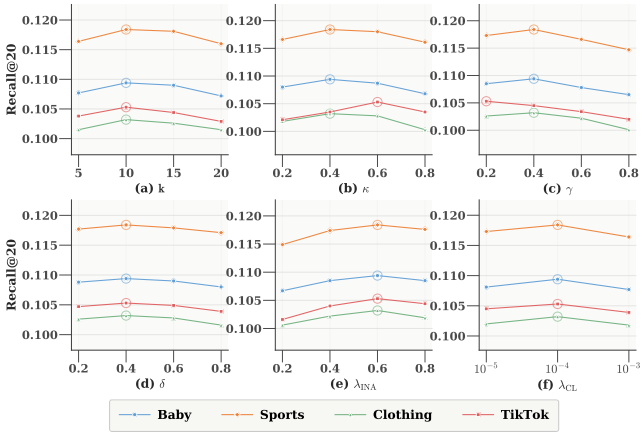


Figure 4: Hyper-parameter analysis for IIMRec across all datasets in terms of Recall@20.

D Baselines

We compare IIMRec against a diverse set of representative baselines, categorized into two groups.

1) General recommendation models:

- **LightGCN** [6] simplifies graph convolutional network (GCN)-based collaborative filtering by eliminating redundant components such as feature transformation and nonlinear activation.
- **SimGCL** [31] performs graph contrastive learning by injecting random noise directly into the representation space instead of augmenting the graph structure.
- **LayerGCN** [35] introduces residual connections into GCN layers to mitigate the over-smoothing problem in LightGCN.
- **NLGCL** [28] leverages naturally occurring high-quality contrastive signals on graphs to enhance recommendation accuracy.

2) Multimodal recommendation models:

- **MMGCN** [20] employs separate GCNs to process and integrate information from different modalities.
- **DualGNN** [17] constructs an auxiliary user-user graph to capture user preference correlations alongside multimodal item features.
- **LATTICE** [32] builds item-item semantic graphs from multimodal features to capture latent correlative signals among items.
- **SLMRec** [16] applies self-supervised learning to multimodal recommendation via feature-level noise perturbation and cross-modal pattern discovery.
- **BM3** [37] simplifies self-supervised multimodal recommendation by generating contrastive views through dropout-based perturbation.
- **MMSSL** [18] employs modality-aware adversarial perturbations for interactive structure learning and combines cross-modal contrastive learning to disentangle shared and modality-specific features.
- **FREEDOM** [36] denoises the user-item interaction graph and constructs a frozen item-item graph from raw modality features.
- **LGMRec** [4] utilizes hypergraph structures to capture both local topological information and global collaborative embeddings.
- **DiffMM** [7] integrates a modality-aware graph diffusion model with cross-modal contrastive learning to improve user representation learning.
- **SMORE** [12] mitigates modality noise by leveraging discriminative spectral properties in the frequency domain.
- **HPMRec** [2] enriches feature diversity and bridges semantic gaps across modalities through hypercomplex representations.
- **MENTOR** [26] designs multi-level cross-modal alignment tasks to enhance the quality of learned representations.
- **COHESION** [23] introduces a dual-stage fusion mechanism to fully exploit composite graph structures.

E Algorithm

The complete training procedure is summarized in Algorithm 1.

E.1 Hyper-parameter Analysis

We investigate the sensitivity of IIMRec to six key hyper-parameters by varying each while fixing the others at their optimal values, and report Recall@20 across all four datasets in Figure 4.

KNN neighbor count k. Performance peaks at k=10 across all datasets and degrades at both extremes: too few neighbors limit the semantic coverage of the item-item graph, while too many introduce noisy edges that dilute the propagated signal.

NCER reweighting strength κ. A moderate κ=0.4 yields the best results on three datasets, with TikTok favoring κ=0.6. Larger values over-amplify high-overlap edges at the expense of useful but structurally weaker connections.

Content-guided UI balance γ. All datasets favor $\gamma \in \{0.2, 0.4\}$, indicating that a modest injection of virtual interactions benefits collaborative propagation, but excessive reliance on the content-guided graph introduces noise.

INA discount factor δ. Performance is relatively stable across $\delta \in [0.2, 0.6]$ and declines at δ=0.8, confirming that soft positives should be discounted to avoid over-weighting uncertain supervision.

INA loss weight λ_{INA}. The optimal value is λ_{INA}=0.6 across all datasets. Too small a weight under-utilizes the augmented signal,

while too large a weight lets soft-positive gradients dominate the primary BPR objective.

Contrastive loss weight $\lambda_{\mathbf{CL}}$. λ_{CL}=10^{-4} is consistently optimal. A larger weight (10^{-3}) forces excessive cross-modal alignment that conflicts with the collaborative filtering signal.